

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000788	Version:	3.0, CURRENT
	Status:	Effective	Effective Date:	25-May-2011
Title:	SM DG57. Troubleshooting.			Page 2 of 8

DG-57 Service Manual

TROUBLESHOOTING

Procedure description
This document is a troubleshooting of the main incidences that can be found in the instrument. <i>Note: Document REGD-0000788_v3 replaces document REGD-0000788_v2.</i>

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000788	Version:	3.0, CURRENT
	Status:	Effective	Effective Date:	25-May-2011
Title:	SM DG57. Troubleshooting.			Page 3 of 8

INITIALIZING INCIDENTS

SAMPLE & REAGENT CAROUSEL INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
No home detected in sample carousel.	Possibly some dirt under the crown.	Clean de crown	
	Faulty home sensor.	Replace the Home Sensor	
False home detection (samples and/or reagents).	Possibly some dirt under the crown or carousel.	Clean de crown and Carousel	
Scratches on reagent labels.	Due to imperfections in the top edge of the vials placed in the carousel.	Replace vials	

REAGENT CAMARA INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
Barcode not read.	Check camera distance (41mm).	Adjust camera orientation and optics via technical service software.	REGD-0000811
	Reagent spinner not working properly	Check motor, cables, ...	REGD-0000797
	Reagent label damaged or out of specifications	Replace the label	
No image (black image) coming from camera.	Camera broken.	Replace the camera	REGD-0000827
	24V to 8V PCA broken (probably only its fuse).	Replace the Fuse or Replace the PCA	
	Matrox Card not installed properly.	Reseat the Matrox Card	REGD-0000779, REGD-0000780
	Video connection cable faulty	Replace the Video cable	

SAMPLE CAMERA INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
Barcode not read	Check camera distance (148mm).	Adjust camera orientation and optics via technical service software.	REGD-0000810
	Barcodes out of specs or badly positioned.	Replace the Barcodes	
Tube width not detected or not detected properly	Check camera distance (148mm).	Adjust camera orientation and optics via technical service software	REGD-0000810
	Check that the distance stated in technical service software is right	Adjust camera orientation and optics via technical service software	REGD-0000810
	Tubes not straight	Check tube springs so the sample tubes are pressed outwards	

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000788	Version:	3.0, CURRENT
	Status:	Effective	Effective Date:	25-May-2011
Title:	SM DG57. Troubleshooting.			Page 4 of 8

No image (black image) coming from camera	Camera broken.	Replace the camera	REGD-0000827
	24V to 8V PCB broken (probably only its fuse).	Replace the Fuse or Replace the PCA	
	Matrox card not installed properly.	Reseat the Matrox Card	REGD-0000779, REGD-0000780
	Connection cable faulty	Replace the Video cable	

HEATING BLOCK INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
No heating	Heater PCA failure	Check heating PCB.	REGD-0000800, REGD-0000821
	24 Power supply failure	Check power section in I/O PCB	REGD-0000800
No cooling	Peltier cells failure	Check Peltier cells.	REGD-0000821
	24V Power supply failure	Check power section in I/O PCB	REGD-0000800
No temperature regulation	Possible sensor broken.	Replace Heater PCA	
	Temperature of the room too high	Ensure about room conditions	
	Wrong adjustment of temperature regulation	Readjust temperature	REGD-0000799

READER CAMERA INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
No image (black image) coming from camera	Camera broken.	Replace the camera	REGD-0000827
	24V to 8V PCB broken (probably only its fuse).	Replace the Fuse or Replace the PCA	
	Matrox card not installed properly.	Reseat the Matrox Card	REGD-0000779, REGD-0000780
	Connection cable faulty.	Replace the Video cable	
No Barcode read	Camera misadjusted	Check camera adjustments.	REGD-0000812
	Card tilted	Check if card is tilted due to wrong gripper adjustment	REGD-0000807, REGD-0000808
"Mil.dll message... file access error" when trying to read a card	The Reference Image has not been created	Create reference image	REGD-0000814

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000788	Version:	3.0, CURRENT
	Status:	Effective	Effective Date:	25-May-2011
Title:	SM DG57. Troubleshooting.			Page 5 of 8

PROBE INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
Level not detected properly	Level Detection PCB damaged.	Replace Level Detection PCA	
	Cable broken / connector faulty in Z chain.	Replace cable	REGD-0000823
	Tube diameter detected badly.	Check camera adjustment.	REGD-0000810
	Dirt under Level Detection PCB.	Clean connection pad under Level detection PCA	
	Bubbles on the liquid	Ensure there are no bubbles on the liquid.	
	Drops in the tube walls	Ensure there are no drops at the tube walls	
	Bent probe	Replace probe	REGD-0000828
Level detected too early	Probably due to probe being wet.	Clean probe. Check for leaks	
	Possible leak in washing station	Check tubing and racors	
	Springs of sample carousel not pressing samples tubes outwards	Check springs tightness	
Pipetting position with reagents too close to neck of vials.	Offset position not adjusted correctly	Adjust reagent position offset in technical service software	REGD-0000796
Liquid leaking on top of card foil.	Some bubbles are considered normal due to pressurised cards.	Change probe	REGD-0000828
Level not detected error when starting probe movement	Wrong adjustment of robot position in technical service software	Readjust probe height in incubator	REGD-0000807
The probe collides with the bottom of sample tubes	Maximum depth misadjustment	Check 'ProfundidadTubo' register	REGD-0000783
The probe collides with the bottom of reagent vials	Maximum depth misadjustment	Check 'ProfundidadVial' register	REGD-0000783
The probe collides with the bottom of diluent bottles	Maximum depth misadjustment	Check 'ProfundidadDiluyente' register	REGD-0000783
The probe collides with the bottom of plunger tubes	Maximum depth misadjustment	Check 'ProfundidadEmbolo' register	REGD-0000783
Current limit error appears	User error	Train user	
	Slipping pulley/belt	Check tightness / status of pulleys and belts	
	Leadscrew and ball bearings not lubricated properly	Lubricate instrument	REGD-0000841

GRIPPER INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
No card detected	Home detector or connections broken.	Replace the Home detector	
	Gripper adjustment wrong (increase Z position).	Adjust the gripper from the service software	REGD-0000807, REGD-0000808
	Pin bent	Replace the bent pin	REGD-0000826
Releasing cards.	Wrong gripper adjustment.	Adjust the gripper from the service software	REGD-0000807, REGD-0000808

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000788	Version:	3.0, CURRENT
	Status:	Effective	Effective Date:	25-May-2011
Title:	SM DG57. Troubleshooting.			Page 6 of 8

Cards tilted.	Gripper pins bent or springs not equal	Replace pins / springs	REGD-0000826
	Gripper not aligned.	Align and adjust gripper	REGD-0000807
	Gripper wrong adjusted.	Adjust the gripper from the service software	REGD-0000807, REGD-0000808
Current limit error appears	User error	Train user	
	Slipping pulley/belt	Check tightness / status of pulleys and belts	
	Leadscrew and ball bearings not lubricated properly	Lubricate instrument	REGD-0000841

FLUID SYSTEM INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
The system is not fed (primed)	Dilutor not initialised	Check the dilutor is initialised	
	Dilutor valve failure	Check that the Dilutor valve turns to input and output	
	Leak in tube or racor	Check tubing and racors	
	Electrovalve not working correctly	Check electrovalves	REGD-0000821
Leak in washing station.	Not enough vacuum.	Check pumps, silencers, racors	
	Tubes/racords obstructed by dirt / squeezed.	Replace tubing, racors	REGD-0000822
	Is final wash performed daily?		
	Home position of probe too low. Probe bent.	Check home position. Replace probe.	REGD-0000828
	Pressure regulator misadjusted (too much pressure)	Readjust pressure regulator	REGD-0000816
	EVE or EVD defective	Replace EVE or EVD	REGD-0000830
	Defective CPC connector	Replace connector	
	Waste container/cap (seal) defective	Change container/cap(seal)	
	Clot in output of wash station	Remove clot	
	Wash station scratched	Replace wash station	
Lack of pressure / vacuum	Probe partially blocked	Replace probe	REGD-0000828
	Pumps problem.	Replace pumps	REGD-0000829
	Tubes obstructed or bent.	Replace damaged tube	REGD-0000822
	Tubes connected wrongly.	Reconnect tubes	
	I/O board broken	Replace I/O PCA	REGD-0000825
Dirt on washing station.	Probably salt from Saline solution	Clean it with distilled water.	
Electrovalves not working.	Electrovalve broken.	Replace Electrovalve	REGD-0000830
	I/O board broken.	Replace I/O board	REGD-0000825
Bubbles in sample line	EVB defective	Replace EVB	REGD-0000830
	Syringe tube connector (silicon tube side) defective or wrongly assembled	Replace/fix tube connector	
	O-ring (inside syringe valve) lost or defective	Replace O-ring	
	Syringe valve defective	Replace valve	
	Syringe piston seal worn out	Replace seal	
	Syringe glass worn out	Replace glass	

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000788	Version:	3.0, CURRENT
	Status:	Effective	Effective Date:	25-May-2011
Title:	SM DG57. Troubleshooting.			Page 7 of 8

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
Overflow in predilution cup	Syringe not properly installed	Install correctly the syringe	
	EVE defective	Replace EVE	REGD-0000830
	Clot expelled during pipetting in the bottom of the cup	Remove clot	
	Silicon tube blocked or squeezed	Replace/fix silicon tube	
	Vacuum failure	Refer to lack of pressure / vacuum section in this table	
Air filter full of water	EVD defective	Replace EVD	REGD-0000830
	Tube connector (bottom filter cup) blocked or defective	Replace tube connector	
	PVC tube blocked or squeezed	Replace/fix tube	
	Fluidics distributor blocked	Replace/fix distributor	

4.9. POWER SUPPLY INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
No signal coming from any camera.	24 to 12V PCB broken (probably fuse).	Replace the Fuse or the PCA	
No carousels movement.	Fuse from Carousels PCB broken.	Replace Fuse	
No Robot movements.	Fuse from Robot PCB broken.	Replace Fuse	
No electrovalves working, no heating, no fanning, door not opening, ...	Fuse from I/O PCB broken.	Replace Fuse	
No centrifuge movement.	Fuse from Centrifuge PCB broken.	Replace Fuse	
Power supply not reaching 24V.	Power supply broken. Sense panduit connector not connected.	Replace Power supply unit.	

4.10. CAVRO DILUTOR INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
Air entering 120° valve.	Connectors not screwed in properly.	Screwed in by hand only, no need for tools.	
Liquid leak.	Tube connectors broken or badly connected.	Replace connector or connected properly	
	Syringe tip broken.	Replace syringe	REGD-0000831

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000788	Version:	3.0, CURRENT
	Status:	Effective	Effective Date:	25-May-2011
Title:	SM DG57. Troubleshooting.			Page 8 of 8

4.11. ELECTROLUMINESCENT SCREEN INCIDENTS

Symptom	Possible Cause	Recommended Action	PROCEDURE NUMBER
Not switching on.	Not properly connected	Check connector under reagents carousel.	
	Screen broken	Replace Screen	
	PCA reagents carousel broken	Replace PCA	REGD-0000825

DOCUMENT CHANGE HISTORY

Version	Date	Change	Rationale
2.0	2005/05/09	<i>Document REGD-0000788_v3 replaces document 57016VOM000.</i>	Document referente changed.
3.0	See document header	<i>Document REGD-0000788_v3 replaces document REGD-0000788_v2.</i>	Document version changed.
3.0	See document header	Document title.	Commercial reference substituted by project reference.
3.0	See document header	Header title.	Increase information to easy document search.